

Mathematics Challenge Paper

Time: 1 hour

Total Marks: 100

Name: _____ Year: _____

Part A – Short Questions (1 mark each)

1. Round 43,697 to the nearest thousand.
2. What is the value of the 6 in 768,204?
3. Simplify the fraction $\frac{18}{24}$.
4. Convert 3.6 km into meters.
5. Multiply: 1.25×4
6. Divide: $144 \div 12$
7. Write 0.75 as a fraction in simplest form.
8. Write the next two numbers: 3, 6, 12, 24, __, __
9. Find the perimeter of a square with side 7 cm.
10. Add: $3.45 + 2.70 + 1.85$
11. Subtract: $5000 - 2738$
12. Multiply: 36×8
13. Divide: $315 \div 9$
14. Convert 1250 g to kilograms.
15. Round 86.72 to the nearest whole number.

Part B – Reasoning and Application (3 marks each)

16. A fruit seller has 180 apples. He packs them into boxes of 12. How many boxes can he make? How many apples left?
17. A rectangle is 12 cm long and 8 cm wide. Find its perimeter and area.
18. A 3.6 m rope is cut into 5 equal parts. How long is one part?
19. A bag costs \$40. If there's a 25% discount, what is the discounted price?
20. Find the mean of: 12, 18, 22, 16, 14
21. A water tank holds 15.5 L. How many 2.5 L bottles are needed to fill it?

22. Plot points A(2,3), B(4,1), C(2,1) and name the shape formed.
23. A garden is 12 m long and 7 m wide. A 5 m square patio is placed inside. Find remaining area.
24. Find the missing number: $_ + 219 = 842$
25. There are 5 red, 3 blue, and 2 green marbles. What is the probability of picking a red one?

Part C – Word Problems (5 marks each)

26. Sam saves \$75 every week. How many weeks to save \$1200? How much in 17 weeks?
27. A class has 128 students. Each gets 3 books. Books are packed in boxes of 24. How many boxes?
28. A tile is 20 cm $\times$ 30 cm. How many tiles to cover a 3 m $\times$ 4 m floor?
29. Class collects 1.8 kg of paper daily. How much in 4 weeks (5 days/week)?
30. A family uses 6.5 L of milk a week. How much in 8 weeks?
31. A car travels 48 km in 1 hour. How far in 6 hours?
32. A rectangle has length 9.5 m, width 4.2 m. Find perimeter and area.
33. A shop had 125 pens. 38 were sold in the morning, 27 in the afternoon. How many left?
34. A pie chart shows time spent: 6 hrs sleep, 2 hrs homework, 1 hr TV, 1 hr games. What fraction is sleep?
35. Jane read 72 pages in 3 days. If she reads 24 pages/day, how many more days?

Extra Challenge (Bonus)

36. Write an equation for: “5 more than twice a number is 17”. Solve it.
37. Draw a bar chart to show: Apples - 10, Oranges - 8, Bananas - 5, Grapes - 7.
38. Write 3 equivalent fractions for $\frac{2}{3}$.
39. List all prime numbers between 10 and 30.
40. Write all factors of 36.

End of Paper